

MK-UW: Metric Keypoints for Underwater Pose Estimation

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Abstract: Underwater pose estimation remains challenging due to severe image degradation and the scarcity of labeled underwater data. Motivated by the strong performance of metric-keypoint techniques in in-air applications, we propose MK-UW (Metric-Keypoint-based Underwater Pose Estimation), a two-stage transfer learning framework that adapts a geometry-aware metric-keypoint correspondence model from in-air to underwater environments. The framework first performs supervised fine-tuning on synthetic underwater images with preserved ground-truth poses, followed by self-supervised adaptation on real unlabeled underwater images to reduce the remaining domain gap. Experiments on an Aqualoc-based underwater MapFree benchmark demonstrate that MK-UW consistently outperforms direct transfer and representative classical and learning-based baselines.

Keywords: Underwater pose estimation, Transfer learning, Domain adaptation, Self-supervised learning

1 Introduction

Underwater visual pose estimation is important for robotics applications such as navigation, inspection, and mapping [1, 2]. Although learning-based visual pose estimation methods, such as SuperPoint [3] and MicKey [4], achieve strong performance in in-air environments, direct transfer to underwater environments remains unreliable.

The main challenges arise from severe underwater image degradation caused by absorption, scattering, turbidity, and illumination changes [5, 6], together with the scarcity of labelled underwater pose datasets [1]. Consequently, correspondence models trained on in-air datasets often experience substantial performance degradation underwater. Existing underwater pose estimation methods mainly improve feature robustness through image enhancement or appearance-based feature learning [7, 8], while relatively limited attention has been given to transferring geometry-aware correspondence learning frameworks to underwater environments.

To address this problem, this paper proposes metric keypoints for underwater pose estimation (MK-UW), a two-stage transfer learning framework for adapting geometry-aware metric-keypoint correspondence models from in-air to underwater environments for map-free relative pose estimation without requiring labeled underwater pose datasets. Stage I performs supervised transfer learning on synthetic underwater image pairs generated from labeled RGB-D datasets through underwater image formation models. Stage II further adapts the model to real underwater images using self-supervised learning based on homography consistency and knowledge distillation. In addition, we apply an underwater MapFree-style benchmark [9] based on the Aqualoc dataset [10] and demonstrate improved underwater correspondence reliability and pose estimation accuracy over one-stage transfer and representative baselines.

37 2 Related Work

38 Correspondence-Based Pose Estimation

39 Correspondence-based pose estimation recovers camera motion by detecting and matching visual
40 landmarks and solving geometric constraints such as epipolar geometry or PnP [11, 12]. Classical
41 systems such as PTAM [13] and ORB-SLAM [14, 15, 16] rely on traditional feature descriptors in-
42 cluding Scale-Invariant Feature Transform (SIFT) [17], Speeded-Up Robust Features (SURF) [18],
43 and Oriented FAST and Rotated BRIEF (ORB) [19]. These methods provide efficient feature match-
44 ing but often degrade under severe appearance changes.

45 Learning-based detectors and descriptors such as SuperPoint [20], LF-Net [21], D2-Net [22],
46 R2D2 [23], and DISK [24] improve feature repeatability by learning feature representations directly
47 from training data. Matching networks such as SuperGlue [3] and transformer-based models [25]
48 further improve matching robustness by exploiting global contextual information. More recently,
49 methods such as MicKey [4] jointly learn feature matching and metric pose estimation, making
50 them promising foundations for robust pose estimation under domain shift.

51 Underwater Visual Pose Estimation

52 Due to severe underwater image degradation, early underwater SLAM systems adapted classical
53 feature-based pipelines [1, 26, 2], while later methods incorporated image enhancement or physics-
54 based correction before feature extraction [27, 6, 28]. Recent learning-based approaches [7, 8]
55 improve robustness by learning underwater visual features directly from training data. However,
56 these methods remain constrained by limited labelled underwater data and have limited exploration
57 of learning-based feature matching models for underwater pose estimation.

58 Transfer Learning for Visual Pose Estimation

59 Transfer learning is a natural solution when labeled target-domain data are scarce [29, 30]. Exist-
60 ing domain adaptation methods range from supervised fine-tuning to self-supervised consistency
61 learning [31]. However, most prior work focuses on image classification or semantic understanding
62 tasks [32, 33], rather than feature matching and pose estimation problems that require geometric
63 consistency across views [34]. Motivated by the strong performance of in-air learning-based feature
64 matching models and the scarcity of labeled underwater pose data, this work investigates how such
65 models can be effectively adapted to underwater visual pose estimation through transfer learning.

66 3 Method

67 This section describes the proposed two-stage transfer learning framework for underwater visual
68 pose estimation. The framework progressively adapts a learning-based feature matching model from
69 in-air to underwater environments through supervised transfer on synthetically generated underwa-
70 ter images and self-supervised adaptation on real underwater data. The key objective is to learn
71 geometrically reliable feature correspondences despite severe underwater image degradation.

72 3.1 Framework Overview

73 We address the problem of domain adaptation using two types of data: synthetic underwater images
74 with pose annotations and real underwater images without labels. Synthetic underwater images are
75 generated from labeled RGB-D datasets using underwater image formation models [35], preserving
76 geometric relationships between image pairs. The proposed framework adopts a two-stage train-
77 ing strategy. Stage I fine-tunes a pre-trained in-air model on synthetic underwater images using
78 geometric supervision, while Stage II further adapts the model on real underwater images using
79 self-supervised learning. The overall framework is illustrated in Fig. 1.

80 **Notation:** Let $\mathbb{R}$ and $\mathbb{Z}$ denote the sets of real numbers and integers, respectively. $I \in \mathbb{R}^{H \times W \times C}$
81 denotes an input image, where $H, W \in \mathbb{Z}$ are the image height and width, and $C \in \mathbb{Z}$ is the
82 number of image channels. A pixel location is represented by $\mathbf{x} \in \Omega$, where $\Omega \subset \mathbb{R}^2$ is the

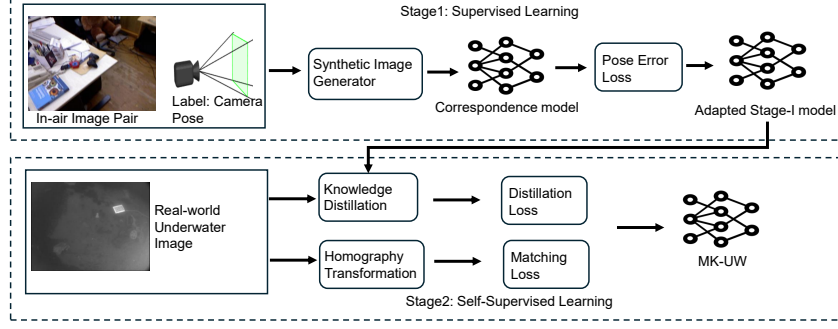


Figure 1: Proposed two-stage metrick-keypoint (MK-UW) transfer framework

image domain. Let $f_\theta(\cdot)$ denote the feature extraction network parameterized by θ . Given an input image, the network produces a keypoint confidence map $S \in [0, 1]^{H \times W}$ and a dense descriptor map $D \in \mathbb{R}^{H \times W \times d}$, where $d \in \mathbb{Z}$ is the descriptor dimension. Given a homography transformation $\mathbf{H} \in \mathbb{R}^{3 \times 3}$, the warping operation applied to images or feature maps is denoted by $\mathcal{W}(\cdot, \mathbf{H})$.

3.2 Stage I: Supervised Learning Using Synthetic Underwater Images

Synthetic Underwater Image Generation In Stage I, supervised learning is performed using synthetic underwater images generated from in-air RGB-D datasets with labeled camera poses. The objective is to adapt the pretrained correspondence model to underwater appearance degradation while preserving the geometric constraints required for pose estimation.

Following the physics-based underwater image formation model in [35], the observed underwater image is formulated as

$$\mathbf{U}_c(\mathbf{x}) = \mathbf{I}_c(\mathbf{x})t_c(\mathbf{x}) + \mathbf{B}_c(\mathbf{x})(1 - t_c(\mathbf{x})) + \eta_c(\mathbf{x}), \quad (1)$$

where $c \in \{R, G, B\}$ denotes the color channel, $\mathbf{U}_c(\mathbf{x})$ denotes the observed underwater image, $\mathbf{I}_c(\mathbf{x})$ is the corresponding latent clear in-air image, $\mathbf{B}_c(\mathbf{x})$ represents the background light, and $\eta_c(\mathbf{x})$ denotes additive noise. The transmission term $t_c(\mathbf{x})$ is defined as

$$t_c(\mathbf{x}) = e^{-\beta_c d(\mathbf{x})}, \quad (2)$$

where $d(\mathbf{x})$ is the scene depth at pixel $\mathbf{x}$ and β_c denotes the wavelength-dependent attenuation coefficient for each color channel.

Using this formulation, underwater-style images are generated from in-air RGB-D datasets such as TUM [36] and KITTI [37] while preserving camera pose information. This enables domain adaptation of the correspondence model under reliable geometric supervision derived from known camera poses. Detailed parameter settings and noise configurations are provided in Appendix A.

Once the synthetic underwater dataset is generated, Stage I fine-tunes the pretrained feature matching model under geometric supervision. The proposed two-stage training framework is model-agnostic and does not depend on a specific backbone architecture. Any pretrained feature matching model that predicts matching-related quantities, such as SuperPoint [20] or R2D2 [23], can be adapted in Stage I and further refined in Stage II.

Stage I: Training Using the Generated Synthetic Dataset Once the synthetic underwater dataset is generated, the correspondence model is initialized with pretrained weights using in-air images and fine-tuned using synthetic underwater image pairs with known relative poses. Since the synthetic images preserve geometric annotations from the original RGB-D datasets, supervised training can be directly applied under reliable geometric constraints. The training objective encourages pose-informative correspondences rather than only adapting to underwater appearance.

Given a pair of synthetic underwater images I and I' with ground-truth relative pose T_{GT} , the correspondence model f_θ predicts a set of correspondences J between the two images. These cor-

116 responses are then passed to a pose solver $h(\cdot)$, which estimates the relative pose. Stage I loss is
 117 defined as

$$\mathcal{L}_{\text{stage1}} = \lambda_{\text{pose}}\mathcal{L}_{\text{pose}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}. \quad (3)$$

118 Here, $\mathcal{L}_{\text{pose}}$ penalizes the difference between the estimated relative pose and the ground-truth relative
 119 pose T_{GT} , while $\mathcal{L}_{\text{reg}}$ regularizes the predicted correspondences to maintain reliable and geometri-
 120 cally consistent matching. The weights λ_{pose} and λ_{reg} balance pose accuracy and correspondence
 121 reliability. The concrete definitions used in our implementation are provided in Sec. 4.1.

122 After Stage I training, the model adapts to underwater appearance degradation while preserving
 123 geometric consistency for pose estimation. This corresponds to the upper half of Fig. 1.

124 3.3 Stage II: Self-supervised Learning Using Real-world Underwater Images

125 Although Stage I adapts the model to underwater appearance using synthetic data, a synthetic-to-
 126 real gap remains. To improve real-world generalization, Stage II further refines the model using
 127 unlabeled underwater images in a self-supervised manner. The Stage I trained model is used as the
 128 teacher model in Stage II to preserve the geometry-aware correspondence knowledge learned from
 129 synthetic supervised transfer.

130 Geometric consistency across multiple views is used as supervision without requiring pose anno-
 131 tations. Homography transformations enforce consistent keypoint detection [20], while descriptor
 132 consistency preserves reliable feature representations learned in Stage I.

133 **Stage II: Self-supervised Adaptation** The proposed self-supervised adaptation stage consists of
 134 four complementary objectives: a detection consistency loss $\mathcal{L}_{\text{det}}$ that encourages repeatable key-
 135 point detection under homography transformations, a descriptor consistency loss $\mathcal{L}_{\text{desc}}$ that pre-
 136 serves descriptor similarity for corresponding points while separating non-corresponding features, a
 137 sparsity regularization term $\mathcal{L}_{\text{spar}}$ that prevents excessive keypoint activation, and a confidence regu-
 138 larization term $\mathcal{L}_{\text{ent}}$ that encourages stable and discriminative keypoint responses. The overall Stage
 139 II objective is defined as

$$\mathcal{L}_{\text{stage2}} = \lambda_1\mathcal{L}_{\text{det}} + \lambda_2\mathcal{L}_{\text{desc}} + \lambda_3\mathcal{L}_{\text{spar}} + \lambda_4\mathcal{L}_{\text{ent}}, \quad (4)$$

140 where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are non-negative weighting coefficients. Together, these objectives encour-
 141 age geometrically consistent, discriminative, and spatially stable feature representations without
 142 requiring ground-truth pose annotations. Detailed Stage II equations and implementation settings
 143 are provided in Sec. 4. Through this self-supervised adaptation process, the model becomes more
 144 robust to real underwater appearance variations while preserving reliable feature matching for pose
 145 estimation. This process corresponds to the lower half of Fig. 1.

146 4 Implementation

147 This section presents the implementation details of the proposed two-stage training framework, in-
 148 cluding dataset preparation, training objectives, and self-supervised adaptation strategies.

149 4.1 Stage I: Supervised Training

150 For Stage I, synthetic underwater data are generated from the TUM RGB-D dataset [36]. Sequential
 151 image pairs with sufficient viewpoint variation are sampled and reorganized into a map-free format,
 152 enabling supervised training of pairwise relative pose estimation with geometric consistency.

153 **Teacher Model Selection:** We pick MicKey [4] as the teacher model for Stage I adaptation due
 154 to its strong in-air pose estimation performance and differentiable correspondence-to-pose pipeline,
 155 which is well aligned with the objective of our correspondence-based underwater pose estimation.
 156 MicKey is trained on the MapFree dataset [9] with four output heads (detection, descriptor, depth

and confidence). Detailed architecture definitions are provided in Appendix B. Based on this specific teacher model, we can construct our detailed loss function.

Stage I Loss: The Stage I objective is defined in (3), where the two terms are

$$\mathcal{L}_{\text{pose}} = \mathbb{E}_{J \sim P(J)} \left[\ell_{\text{pose}}(h(J), T_{\text{GT}}) \right], \quad (5)$$

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{J \sim P(J)} \left[\ell_{\text{reg}}^{\text{corr}}(J) \right]. \quad (6)$$

Here, $\ell_{\text{pose}}(\cdot)$ measures the pose estimation error relative to the ground-truth pose T_{GT} , while $\ell_{\text{reg}}^{\text{corr}}(\cdot)$ regularizes correspondence quality, for example through confidence calibration and matching consistency [3, 38]. Detailed definitions and implementation details are provided in Appendix C.

For a candidate feature pair $(\mathbf{x}_i, \mathbf{x}_j)$ with $\mathbf{x}_i \in I$ and $\mathbf{x}_j \in I'$, the correspondence probability is

$$P(\mathbf{x}_i, \mathbf{x}_j) = P_I(\mathbf{x}_i) \cdot P_{I \rightarrow I'}(\mathbf{x}_j | \mathbf{x}_i) \cdot P_{I'}(\mathbf{x}_j) \cdot P_{I \leftarrow I'}(\mathbf{x}_i | \mathbf{x}_j),$$

where P_I and $P_{I'}$ are keypoint selection probabilities, and $P_{I \rightarrow I'}$ and $P_{I \leftarrow I'}$ are descriptor-based matching probabilities in the forward and backward directions. A procrustes-based solver [39] with RANSAC is used to estimate the relative pose.

4.2 Stage II: Self-supervised Adaptation

For self-supervised learning, we use the Sea-Thru dataset [6] as a real-world underwater dataset containing diverse scenes with severe visual degradation. Since accurate pose annotations are unavailable, Stage II adopts a self-supervised strategy based on geometric consistency across transformed views, enabling further adaptation to real underwater appearance without labelled pose data.

Stage II Loss:

Stage II training follows the homography-consistency and descriptor distillation framework introduced in Sec. 3.3. Given an input image I , a random homography $\mathbf{H}$ generates a warped image

$$I' = \mathcal{W}(I, \mathbf{H}), \quad (7)$$

where $\mathcal{W}(\cdot, \mathbf{H})$ denotes geometric warping. Since warping may introduce invalid image regions, we define a binary validity mask $M(\mathbf{x})$ and the corresponding valid spatial set

$$\Omega_M = \{\mathbf{x} \mid M(\mathbf{x}) = 1\}. \quad (8)$$

Let $f(\cdot)$ denote the feature extraction network that predicts a detection map and descriptor map. For the original and warped images,

$$(S, D) = f(I), \quad (S', D') = f(I'). \quad (9)$$

Warping the predictions from the original image gives

$$\tilde{S} = \mathcal{W}(S, \mathbf{H}), \quad \tilde{D} = \mathcal{W}(D, \mathbf{H}). \quad (10)$$

The detection consistency term encourages repeatable keypoint detection under geometric transformations, similar to SuperPoint [20],

$$\mathcal{L}_{\text{det}} = \frac{1}{|\Omega_M|} \sum_{\mathbf{x} \in \Omega_M} |S'(\mathbf{x}) - \tilde{S}(\mathbf{x})|, \quad (11)$$

where $|\Omega_M|$ denotes the number of valid spatial locations. For descriptor consistency, high-confidence locations are selected as

$$\Omega_\gamma = \{\mathbf{x} \mid S(\mathbf{x}) > \gamma\}, \quad (12)$$

where $\gamma > 0$ is the confidence threshold. Descriptor learning is optimized using a contrastive objective inspired by SuperGlue [3] and MoCo [40],

$$\mathcal{L}_{\text{desc}} = -\frac{1}{|\Omega_\gamma|} \sum_{\mathbf{x} \in \Omega_\gamma} \log \frac{\exp(\text{sim}(D'(\mathbf{x}), \tilde{D}(\mathbf{x}))/\tau)}{\sum_{\mathbf{y} \in \mathcal{N}^-(\mathbf{x})} \exp(\text{sim}(D'(\mathbf{x}), \tilde{D}(\mathbf{y}))/\tau)}, \quad (13)$$

where $\mathcal{N}^-(\mathbf{x})$ denotes the negative sample set and $\tau > 0$ is the temperature parameter. To regularize keypoint distribution and confidence, sparsity and entropy regularization terms are introduced:

$$\mathcal{L}_{\text{spar}} = \left(\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} S(\mathbf{x}) - \rho \right)^2, \quad (14)$$

$$\mathcal{L}_{\text{ent}} = -\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} \left[S(\mathbf{x}) \log S(\mathbf{x}) + (1 - S(\mathbf{x})) \log(1 - S(\mathbf{x})) \right], \quad (15)$$

where ρ controls the desired keypoint density. The final Stage II objective is defined in Eq. (4). The settings of λ_1 , λ_2 , λ_3 , and λ_4 are provided in Appendix F, while detailed training pipelines for both stages are provided in Appendix D.

For clarity, the above losses are defined for a single image pair. During training, the objective is averaged over a mini-batch B of real underwater image pairs:

$$\mathcal{L}_{\text{Stage II}} = \frac{1}{B} \sum_{i=1}^B \left(\lambda_1 \mathcal{L}_{\text{det}}^{(i)} + \lambda_2 \mathcal{L}_{\text{desc}}^{(i)} + \lambda_3 \mathcal{L}_{\text{spar}}^{(i)} + \lambda_4 \mathcal{L}_{\text{ent}}^{(i)} \right), \quad (16)$$

5 Experiment Results

This section evaluates the proposed MK-UW through qualitative comparisons, quantitative benchmarks, and ablation studies against both classical and learning-based baselines. Additional details are provided in the Appendices.

5.1 Qualitative Results

This subsection qualitatively compares feature matching robustness and correspondence consistency under severe underwater image degradation. The comparisons are conducted on multiple underwater datasets, including the grayscale Aqualoc dataset [10] and the RGB Sea-Thru dataset [6]. The Sea-Thru dataset is used only for evaluation and is not included during training. Representative examples are shown in Fig. 2. From left to right, the compared methods are ORB [19], Scale-Invariant Feature Transform (SIFT) [17], SuperPoint [20], UFEN [7], the original MicKey model [4], and MK-UW.

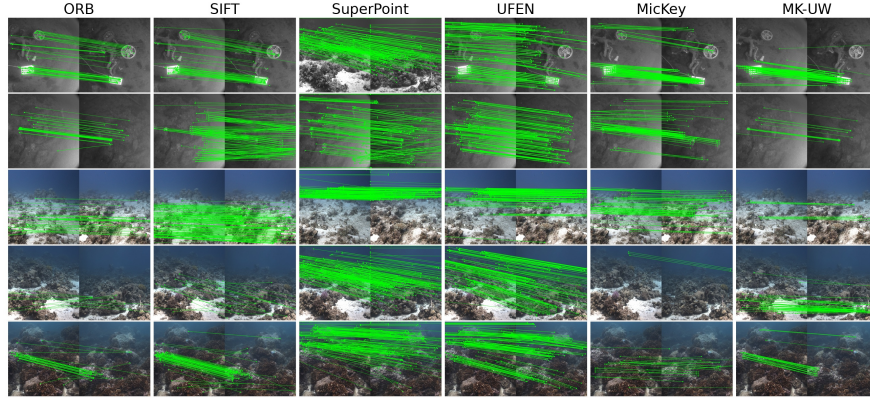


Figure 2: Qualitative comparison of feature matching results on challenging underwater scenes from the Aqualoc [10] (row 1 and 2) and Sea-Thru [6] (row 3 - 5) datasets.

Fig. 2 compares feature matching performance under challenging underwater conditions. Classical methods such as ORB and SIFT often produce many spatially scattered and geometrically inconsistent matches, which become unstable under blur, scattering, low contrast, and illumination variation. Learning-based methods such as SuperPoint and UFEN improve robustness, but many correspondences are still distributed over visually ambiguous or low-texture regions, limiting pose reliabil-

ity. In contrast, the proposed MK-UW generates fewer but more reliable correspondences concentrated on stable object-level structures such as boundaries, corners, and textured regions. Although sparse, these matches provide stronger geometric constraints for relative pose estimation. This behavior originates from the metric-keypoint formulation inherited from MicKey, which prioritizes pose-informative correspondences. Through the proposed underwater transfer learning framework, MK-UW preserves this correspondence selection behavior while adapting to underwater appearance degradation, resulting in more stable matching and improved pose estimation performance.

5.2 Quantitative Results

This subsection quantitatively evaluates the proposed MK-UW framework on underwater map-free relative pose estimation. The objective is to compare the proposed method against classical, CNN-based, transformer-based, and geometry-aware baselines under challenging underwater conditions.

Table 1 compares MK-UW with classical and learning-based baselines on an underwater MapFree benchmark [9] constructed from the Aqualoc dataset [10]. For ORB, SIFT, SURF, SuperPoint, and UFEN, brute-force matching followed by RANSAC is used for relative pose estimation. For DINO-v2, correspondences are obtained using patch-level descriptor similarity before RANSAC. Detailed metric definitions are provided in Appendix E.

Table 1: Results of Transferred Mickey on Underwater Mapfree

Model (steps)	TranE↓	RotE↓	RepE↓	PrecP↑	AucP↑	PrecV↑	AucV↑
ORB[19]	2.41	27.72	194.42	8.41	20.42	21.77	20.70
SIFT[41]	2.70	29.48	207.73	9.09	19.96	20.45	21.01
SURF[42]	3.52	38.54	231.51	8.16	18.74	18.59	17.42
SuperPoint[20]	2.88	24.47	178.71	10.05	20.51	20.68	22.66
UFEN[7]	2.82	23.10	175.42	9.87	21.93	23.74	23.52
DINO-v2[43]	2.65	26.83	180.45	9.13	21.03	22.53	24.58
Mickey (in-air)	1.42	18.80	171.01	9.97	25.35	29.38	46.42
MK-UW (ours)	1.35	15.06	142.65	13.92	29.30	37.58	57.92

Table 1 shows that MK-UW achieves the best performance across all evaluation metrics. Compared with the original Mickey model, the proposed framework reduces translation, rotation, and reprojection errors while improving pose precision and VCRE success rates, demonstrating improved robustness under underwater domain shift. Compared with classical descriptors (ORB, SIFT, SURF) and learning-based methods (SuperPoint, UFEN, DINO-v2), MK-UW achieves lower geometric errors and higher pose success rates under underwater blur, scattering, illumination variation, and low-contrast conditions. These results demonstrate the effectiveness of the proposed two-stage transfer learning framework for underwater map-free pose estimation.

5.3 Ablation Study

This subsection evaluates the contribution of different stages and training objectives in the proposed MK-UW framework. The complete pipeline consists of two stages: *Stage I*, which performs supervised transfer learning on synthetic underwater image pairs with ground-truth poses, and *Stage II*, which performs self-supervised adaptation using real underwater images. Table 2 compares different Stage I/Stage II combinations and analyzes the influence of individual Stage II objectives. The first row, *S1+S2 (MK-UW)*, corresponds to the final MK-UW configuration used in the previous experiments. Detailed Stage II components, including homography adaptation, knowledge distillation, and sparsity/entropy regularization, are illustrated in the training pipeline shown in Appendix D.

Table 2 shows that *S1+S2 (MK-UW)* achieves the best overall performance. This demonstrates that combining supervised synthetic transfer with self-supervised real-world adaptation provides the most effective underwater transfer strategy. Compared with *S1+S2 (All heads)*, restricting training on the detection and descriptor head yields better performance, because these heads are directly responsible for producing repeatable keypoints and discriminative local descriptors, which are the

Table 2: Ablation study on Aqualoc-based underwater map-free evaluation.

Model	TranE↓	RotE↓	RepE↓	PrecP↑	AucP↑	PrecV↑	AucV↑
S1+S2 (MK-UW)	1.35	15.06	142.65	13.92	29.30	37.58	57.92
S1+S2 (All heads)	1.37	15.20	148.92	13.50	28.78	37.28	56.45
S1+S2 (Homo only)	1.38	14.94	159.74	13.60	24.41	36.24	47.11
S1+S2 (KD only)	1.37	14.63	160.87	13.08	25.68	36.24	48.39
S1+S2 (No Spar & Ent)	1.38	14.80	157.89	13.50	25.26	35.93	47.45
S2 only (No S1)	1.44	20.01	175.68	12.25	32.91	30.42	57.18
S1 only (No S2)	1.38	15.28	150.83	13.29	26.54	33.30	56.13
MicKey (in-air)	1.42	18.80	171.01	9.97	25.35	29.38	46.42

S1: Stage I ; S2: Stage II; MK-UW: knowledge distillation applied only to the detection and descriptor head; All heads: knowledge distillation applied to all MicKey output heads, including detection, descriptor, depth, and confidence heads; Homo: homography adaptation; KD: knowledge distillation; Spar: sparsity loss $\mathcal{L}_{\text{spar}}$; Ent: entropy regularization loss $\mathcal{L}_{\text{ent}}$; In-air: the pretrained MicKey model.

most critical components for correspondence establishment. In contrast, applying distillation to all heads may over-constrain the student model. Both *S1+S2 (Homo only)* and *S1+S2 (KD only)* improve over the original MicKey model but remain weaker than the full framework (two-stage MK-UW), indicating complementary benefits from homography consistency and descriptor distillation. Removing sparsity and entropy regularization further reduces performance. Finally, *S2 only (No S1)* produces substantially larger geometric errors, showing that supervised synthetic transfer provides an important initialization for stable underwater adaptation.

6 Limitations

Several limitations of the proposed MK-UW framework should be noted. First, the current framework focuses on pairwise relative pose estimation and does not explicitly model temporal consistency required for visual odometry or SLAM systems. Second, the synthetic underwater image generation process relies on simplified underwater image formation assumptions and may not fully capture complex real underwater phenomena such as dynamic lighting, severe turbidity, marine particles, and moving objects. Third, the current evaluation is mainly conducted on the Aqualoc-based underwater MapFree benchmark, and broader validation on more diverse underwater environments is still required. In addition, the Stage II homography-based self-supervision may become less reliable under strong non-planar scene deformation or large viewpoint changes. Finally, as MK-UW inherits a transformer-based correspondence model, it may require more computation and memory than lightweight hand-crafted features, which could limit deployment on resource-constrained underwater robots.

Future work will investigate underwater visual odometry, multi-view geometric optimization, more realistic underwater simulation, and broader cross-environment adaptation.

7 Conclusions and Discussions

This paper proposed MK-UW, a two-stage transfer learning framework for adapting metric-keypoint correspondence models from in-air to underwater environments for map-free relative pose estimation. Stage I performs supervised transfer on synthetic underwater image pairs, while Stage II applies self-supervised adaptation on real underwater images using homography consistency and descriptor distillation. Experiments on the Aqualoc-based underwater MapFree benchmark show that MK-UW improves correspondence reliability and pose estimation accuracy over direct transfer, classical descriptors, and learning-based baselines. Ablation studies further demonstrate the complementary benefits of supervised synthetic transfer and self-supervised real-world adaptation.

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A Synthetic Underwater Data Generation Setup

This appendix section gives the detailed implementation of synthetic underwater data generation.

For the underwater image formation in Eq. (1) and (2):

$$\mathbf{U}_c(\mathbf{x}) = \mathbf{I}_c(\mathbf{x})t_c(\mathbf{x}) + \mathbf{B}_c(\mathbf{x})(1 - t_c(\mathbf{x})) + \eta_c(\mathbf{x}) \quad (17)$$

$$t_c(\mathbf{x}) = e^{-\beta_c d(\mathbf{x})}, \quad (18)$$

The background light $\mathbf{B}_c$ is defined as [6]

$$\mathbf{B}_c(\mathbf{x}) = \frac{b(\lambda_c)E(d(\mathbf{x}), \lambda_c)}{\beta_c} \quad (19)$$

where $b(\lambda_c)$ denotes the wavelength-dependent scattering coefficient and $E(d(\mathbf{x}), \lambda_c)$ represents the residual light energy at depth $d(\mathbf{x})$ for wavelength λ_c . The residual energy is modeled as

$$E(d(\mathbf{x}), \lambda_c) = E(0, \lambda_c)e^{-K_d(\lambda_c)d(\mathbf{x})} \quad (20)$$

where $E(0, \lambda_c)$ is the light energy at the water surface, $K_d(\lambda_c) > 0$ is the attenuation coefficient of the medium, and $\lambda_c > 0$ denotes the wavelength corresponding to channel c . In this work, we adopt in-air datasets that provide ground-truth camera poses and aligned depth maps as the source of raw images for synthetic generation. Although our experiments use the TUM RGB-D dataset [36] due to its accurate pose annotations and high-quality depth measurements, the proposed generation framework is not restricted to a specific dataset. It can be applied to any in-air dataset with reliable pose and depth information.

The input to our generation model consists of the raw in-air image, its corresponding depth map, and a selected set of parameters β_c , $E(d, \lambda_c)$, and $K_d(\lambda_c)$. The physical parameters are selected using two complementary strategies: (i) random sampling within physically plausible ranges to simulate diverse underwater conditions [6, 44, 45], and (ii) estimation from real underwater images to match specific environments and reduce the domain gap. The output is a set of underwater-style images accompanied by ground-truth camera poses, enabling controlled evaluation of underwater visual algorithms. In this work we picked the latter one and a generated example is shown in Fig. 3.



Figure 3: Examples of synthetic underwater images.

B MicKey Architecture

MicKey [4] is a learning-based keypoint detection and description network designed for metric relative pose estimation. It is trained on in-air image pairs from the MapFree dataset [9]. As shown in Fig. 4, MicKey takes an image as input and extracts shared visual features through a backbone network. These features are then processed by four prediction heads, each responsible for a keypoint-related quantity.

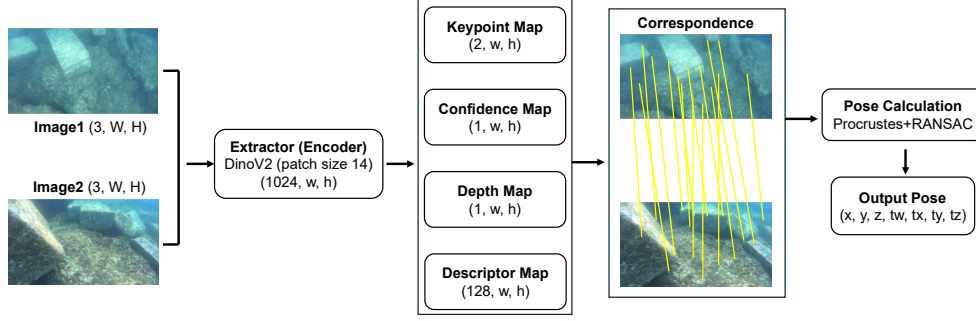


Figure 4: MicKey architecture [4]. The network predicts 2D keypoint offsets $\mathbf{U}$, depth estimates $\mathbf{Z}$, confidence scores $\mathbf{C}$, and 128-dimensional descriptors $\mathbf{D}$, which are used to form probabilistic correspondences and estimate relative pose through a differentiable pose solver.

Specifically, for each image, MicKey predicts 2D keypoint offsets $\mathbf{U}$, depth estimates $\mathbf{Z}$, confidence scores $\mathbf{C}$, and local descriptors $\mathbf{D}$. The 2D offset head $\mathbf{U}$ refines keypoint locations within local image regions. The depth head $\mathbf{Z}$ provides metric depth information associated with predicted keypoints. The confidence head $\mathbf{C}$ estimates the reliability of each predicted keypoint or correspondence candidate. The descriptor head $\mathbf{D}$ outputs 128-dimensional local descriptors used to measure feature similarity across image pairs.

Given two images, the predicted descriptors and confidence scores are used to construct probabilistic correspondences between candidate keypoints. These correspondences, together with the predicted depth and keypoint locations, are passed to a differentiable pose solver to estimate the relative camera pose. This design explicitly links correspondence prediction with metric pose estimation, which makes MicKey suitable as a geometry-aware teacher model for our transfer learning framework.

C Specific loss for stage I

Let the estimated relative pose is $\hat{T} = h(J) = (\hat{\mathbf{R}}, \hat{\mathbf{t}})$, and the ground-truth relative pose is $T_{GT} = (\mathbf{R}_{GT}, \mathbf{t}_{GT})$. Here, $\mathbf{R} \in SO(3)$ denotes a 3×3 relative rotation matrix that describes the orientation change from the source camera frame to the target camera frame, and $\mathbf{t} \in \mathbb{R}^3$ denotes a 3D relative translation vector that describes the displacement between the two camera centers. Accordingly, $\hat{\mathbf{R}}$ and $\hat{\mathbf{t}}$ are the predicted relative rotation and translation, while $\mathbf{R}_{GT}$ and $\mathbf{t}_{GT}$ are their ground-truth counterparts.

The specific pose loss and regularization loss from Eq. (5) and (6) is defined as

$$\ell_{\text{pose}}(\hat{T}, T_{GT}) = \alpha_R d_R(\hat{\mathbf{R}}, \mathbf{R}_{GT}) + \alpha_t d_t(\hat{\mathbf{t}}, \mathbf{t}_{GT}), \quad (21)$$

where α_R and α_t balance the rotation and translation terms. The rotation error is measured by the geodesic distance on $SO(3)$:

$$d_R(\hat{\mathbf{R}}, \mathbf{R}_{GT}) = \arccos \left(\frac{\text{tr}(\hat{\mathbf{R}}^\top \mathbf{R}_{GT}) - 1}{2} \right), \quad (22)$$

where $\text{tr}(\cdot)$ denotes the trace operator, i.e., the sum of the diagonal elements of a square matrix. This term computes the angular difference between the predicted and ground-truth rotations.

The translation error is defined as the angular difference between the predicted and ground-truth translation directions:

$$d_t(\hat{\mathbf{t}}, \mathbf{t}_{GT}) = \arccos \left(\frac{\hat{\mathbf{t}}^\top \mathbf{t}_{GT}}{\|\hat{\mathbf{t}}\|_2 \|\mathbf{t}_{GT}\|_2} \right). \quad (23)$$

Since monocular relative pose estimation usually recovers translation only up to an unknown scale, this term compares translation directions rather than absolute translation magnitudes.

D Training Pipeline

Fig. 5 illustrates the supervised transfer learning pipeline used in Stage I. Given an in-air image pair (I_1, I_2) with camera poses, we compute the ground-truth relative pose and generate the corresponding synthetic underwater image pair (I'_1, I'_2) . The synthetic image pair is then used as input to the fine-tuned model, whose estimated relative pose is supervised by the ground-truth relative pose through the Stage I training loss.

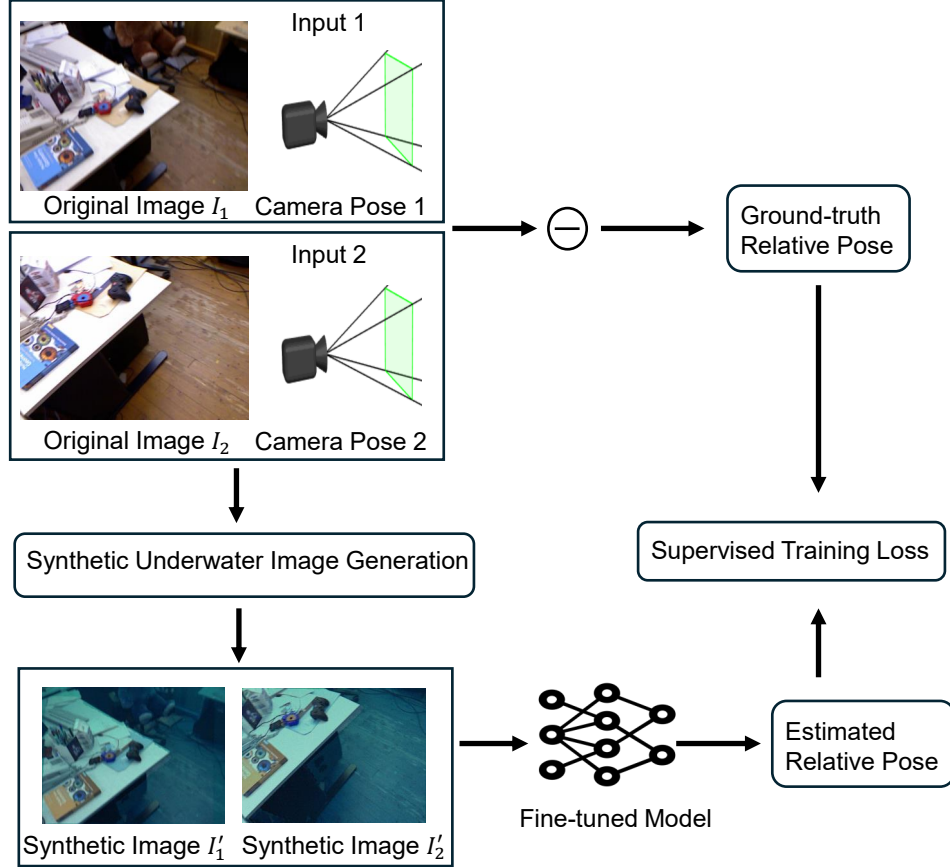


Figure 5: Stage I supervised transfer learning pipeline. In-air image pairs with pose annotations are converted into synthetic underwater image pairs, while the original pose annotations are preserved to supervise relative pose estimation.

Fig. 6 illustrates the self-supervised adaptation pipeline used in Stage II. Given an unlabeled real underwater image, a homography transformation is applied to construct a geometrically related image pair. The student model is trained using feature consistency, descriptor consistency, sparsity, and entropy losses. In addition, knowledge distillation from the teacher model is used to preserve the geometry-aware correspondence representation during adaptation.

E Evaluation Metrics

We follow the map-free relocalization evaluation protocol introduced in [9]. Given a reference image and a query image, the task is to estimate the metric relative pose of the query camera with respect to the reference image. Let the ground-truth query pose be $T = [R \mid \mathbf{t}] \in SE(3)$ and the estimated pose be $\hat{T} = [\hat{R} \mid \hat{\mathbf{t}}] \in SE(3)$, where $R, \hat{R} \in SO(3)$ and $\mathbf{t}, \hat{\mathbf{t}} \in \mathbb{R}^3$. In our evaluation, lower values

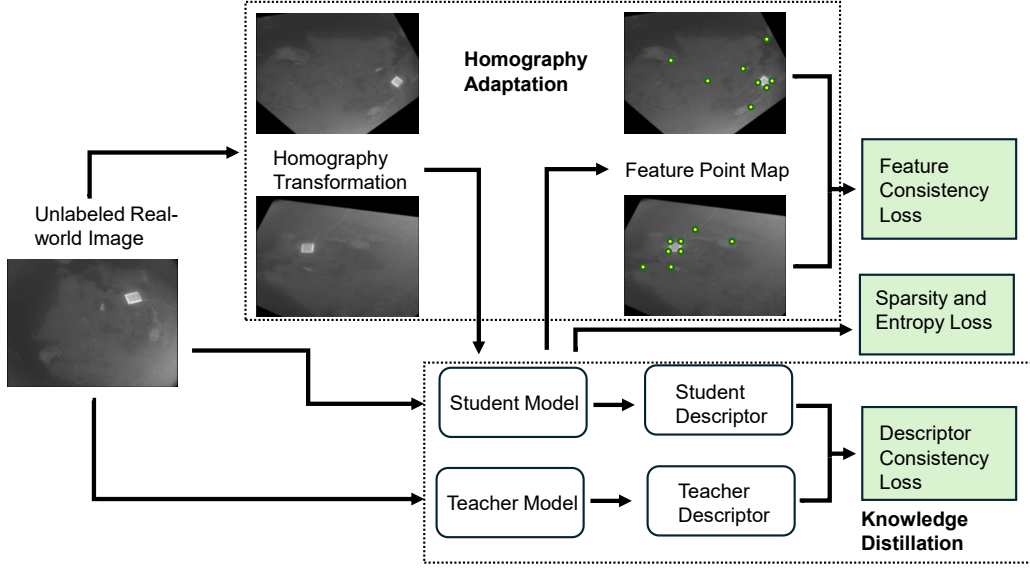


Figure 6: Stage II self-supervised adaptation pipeline. Real underwater images are adapted through homography-based consistency learning and teacher-student knowledge distillation, without requiring ground-truth underwater pose labels.

are better for translation error, rotation error, and reprojection error, while higher values are better for precision and area-under-curve metrics.

Translation error. Translation error measures the Euclidean distance between the estimated and ground-truth camera centers:

$$e_t = \|\hat{\mathbf{c}} - \mathbf{c}\|_2, \quad (24)$$

where $\mathbf{c}$ and $\hat{\mathbf{c}}$ denote the ground-truth and estimated camera centers in the reference coordinate frame, respectively. When poses are represented in camera-to-world form, the camera center is directly given by the translation vector. When poses are represented in world-to-camera form, the camera center is computed as $\mathbf{c} = -R^\top \mathbf{t}$.

Rotation error. Rotation error measures the angular difference between the estimated and ground-truth rotations:

$$e_R = \cos^{-1} \left(\frac{\text{tr}(\hat{R}R^\top) - 1}{2} \right) \cdot \frac{180}{\pi}. \quad (25)$$

The error is reported in degrees. In implementation, the argument of $\cos^{-1}(\cdot)$ is clipped to $[-1, 1]$ for numerical stability.

Virtual Correspondence Reprojection Error. Following [9], we also report Virtual Correspondence Reprojection Error (VCRE), denoted as e_{vcre} . VCRE measures the image-space misalignment induced by the estimated pose. Since the map-free setting does not assume a reconstructed 3D map, a set of virtual 3D points $\mathcal{P} = \{\mathbf{p}_i\}_{i=1}^N$ is placed in the query camera coordinate system. These points are transformed and projected using both the ground-truth and estimated poses. The VCRE is then defined as the average Euclidean distance between the corresponding projected pixels:

$$e_{\text{vcre}} = \frac{1}{N} \sum_{i=1}^N \left\| \pi(KT^{-1}\mathbf{p}_i) - \pi(K\hat{T}^{-1}\mathbf{p}_i) \right\|_2, \quad (26)$$

where K is the camera intrinsic matrix and $\pi(\cdot)$ denotes perspective projection from homogeneous camera coordinates to image coordinates. VCRE is reported in pixels. Intuitively, this metric es-

481 timates how much virtual content would be displaced in the image if rendered using the estimated
482 pose instead of the ground-truth pose.

483 **Pose precision.** Pose precision evaluates the fraction of image pairs whose predicted pose is within
484 predefined translation and rotation thresholds. A pose estimate is considered correct if

$$e_t < \tau_t \quad \text{and} \quad e_R < \tau_R, \quad (27)$$

485 where τ_t and τ_R are translation and rotation thresholds, respectively. In the original MapFree bench-
486 mark implementation, the default thresholds are $\tau_t = 0.25\text{m}$ and $\tau_R = 5^\circ$. The pose precision is
487 computed as

$$\text{Prec}_P = \frac{1}{M} \sum_{m=1}^M \mathbb{I} \left[e_t^{(m)} < \tau_t \wedge e_R^{(m)} < \tau_R \right], \quad (28)$$

488 where M is the number of evaluated image pairs and $\mathbb{I}[\cdot]$ is the indicator function.

489 **VCRE precision.** VCRE precision evaluates the fraction of image pairs whose reprojection error
490 is below a pixel threshold:

$$\text{Prec}_V = \frac{1}{M} \sum_{m=1}^M \mathbb{I} \left[e_{\text{vcre}}^{(m)} < \tau_v \right], \quad (29)$$

491 where τ_v is the VCRE threshold. The MapFree benchmark commonly uses VCRE thresholds corre-
492 sponding to a percentage of the image diagonal, such as 5% or 10%. In the public evaluation code,
493 a default threshold of 90 pixels is used for the single-frame benchmark.

494 **Confidence-aware AUC.** In addition to measuring whether an estimate is correct, the MapFree
495 protocol also evaluates whether a method can assign meaningful confidence to its predictions. Let
496 $s^{(m)}$ be the confidence score of the m -th pose estimate. For a confidence threshold γ , predictions
497 with $s^{(m)} \geq \gamma$ are accepted, while the remaining predictions are rejected. This gives a precision-
498 recall curve:

$$\text{Precision}(\gamma) = \frac{\sum_{m=1}^M \mathbb{I}[s^{(m)} \geq \gamma] \mathbb{I}[\text{correct}^{(m)}]}{\sum_{m=1}^M \mathbb{I}[s^{(m)} \geq \gamma]}, \quad (30)$$

499

$$\text{Recall}(\gamma) = \frac{\sum_{m=1}^M \mathbb{I}[s^{(m)} \geq \gamma]}{M}. \quad (31)$$

500 The area under this confidence-based precision-recall curve is reported as AUC. When the correct-
501 ness condition is defined using Eq. (27), the metric is denoted as AucP. When correctness is defined
502 using $e_{\text{vcre}} < \tau_v$, the metric is denoted as AucV.

503 **Reported metrics.** In the main paper, we report the following metrics:

- 504 • **TranE**: average median translation error e_t .
- 505 • **RotE**: average median rotation error e_R .
- 506 • **RepE**: average median reprojection error, corresponding to VCRE in the MapFree protocol.
- 507 • **PrecP**: precision under the joint translation and rotation pose threshold.
- 508 • **AucP**: confidence-aware AUC under the joint pose threshold.
- 509 • **PrecV**: precision under the VCRE threshold.
- 510 • **AucV**: confidence-aware AUC under the VCRE threshold.

F Training Parameter Setup

The implemented Stage II objective in equation (4) is

$$\mathcal{L}_{\text{stage2}} = \lambda_1 \mathcal{L}_{\text{det}} + \lambda_2 \mathcal{L}_{\text{desc}} + \lambda_3 \mathcal{L}_{\text{spar}} + \lambda_4 \mathcal{L}_{\text{ent}},$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are non-negative loss weights.

For the Stage II objective in Eq. (4), we select the weighting coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ using two practical strategies.

Fixed balancing strategy: We first monitor the raw magnitudes of the four sub-losses during warm-up iterations and set fixed coefficients so that their weighted contributions are of similar scale. For example, we observe that the entropy term $\mathcal{L}_{\text{ent}}$ is often around two orders of magnitude larger than $\mathcal{L}_{\text{det}}$ and $\mathcal{L}_{\text{desc}}$. Therefore, we set λ_4 about 100 times smaller than λ_1 and λ_2 (and similarly tune λ_3) to avoid domination by a single term.

Normalized-gradient strategy: As an alternative, we adapt weights according to gradient magnitude inspired by GradNorm [46]. Let $g_k = \|\nabla_{\theta}(\lambda_k \mathcal{L}_k)\|_2$ for $k \in \{1, 2, 3, 4\}$ and $\bar{g} = \frac{1}{4} \sum_{k=1}^4 g_k$. We update coefficients to equalize gradient scales, for example

$$\lambda_k \leftarrow \lambda_k \left(\frac{\bar{g}}{g_k + \epsilon} \right)^{\alpha},$$

where $\epsilon > 0$ ensures numerical stability and $\alpha \in (0, 1]$ controls update strength. This strategy reduces manual tuning and keeps all sub-objectives active during optimization.

We have done a sensitivity analysis for these parameters. The default parameters for λ is 0.5, 0.5, 0.01, 0.001.

Table 3: Sensitivity analysis of Stage II hyperparameters on Aqualoc-based underwater map-free evaluation.

Parameter	Value	TranE↓	RotE↓	RepE↓	PreP↑	PreV↑	AucP↑	AucV↑
λ_1 (Det)	0.25	1.380	15.14	160.02	13.3	28.2	36.7	55.0
	1.0	1.375	15.01	159.02	13.3	27.8	36.2	54.5
	2.0	1.378	14.84	159.73	13.2	29.6	35.8	54.2
	4.0	1.378	14.83	159.66	13.2	28.7	35.6	53.3
λ_2 (Desc)	0.25	1.384	15.10	157.55	13.5	27.8	35.5	53.1
	1.0	1.380	14.84	159.32	13.5	28.3	36.1	54.4
	2.0	1.381	14.75	159.86	13.6	26.9	38.4	56.2
	4.0	1.361	15.82	172.28	11.9	26.9	29.9	51.2
λ_3 (Spar)	5e-3	1.380	14.79	159.43	13.1	28.6	31.2	54.7
	0.1	1.379	14.88	158.45	13.3	29.3	29.0	49.6
	1.0	1.375	14.89	160.83	13.5	27.6	24.7	45.1
	2.0	1.377	14.62	157.40	13.2	27.3	24.0	46.0
λ_4 (Ent)	5e-4	1.380	14.76	159.55	13.5	28.5	31.6	53.7
	0.01	1.382	14.49	156.82	13.6	28.6	28.4	48.7
	0.1	1.379	15.24	157.43	13.3	29.3	26.2	49.4
	1.0	1.378	15.15	160.24	13.6	28.0	26.9	49.3